Software Test Plan (STP)

> **Project:** Insurance Workflow Automation System
> **Version:** 1.0
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> **Date:** 31-08-2025
> **Status:** Approved

1. Introduction

Purpose: Defines the test plan for the Insurance Workflow Automation System v1.0 — objectives, scope, strategy, resources, schedule, and responsibilities.

Scope: Testing covers quote generation, underwriting, policy issuance, claims processing, workflow orchestration, and notifications. Core actuarial model internals and third-party vendor systems are excluded.

References: IWAS SRS v1.0, SAD v1.0, IRDAI standards, PCI-DSS.

Definitions: IWAS, KYC, BPMN, RTM, SLA, CI/CD, UAT.

2. Test Items

- Quote generation module
- Underwriting / risk scoring module
- Policy issuance & renewal module
- Claims processing module
- Workflow orchestration engine
- Notification service
- Admin & RBAC module

3. Features to be Tested

Feature ID	Feature
IWA-F-001	Generate personalized quote
IWA-F-002	Automated risk scoring
IWA-F-010	Policy issuance with digital PDF
IWA-F-011	Policy renewal reminders
IWA-F-020	Claim registration with document upload
IWA-F-021	BPMN multi-level approval workflow
IWA-F-022	Automated fraud detection
IWA-NF-001	API response time ≤ 2s (95th percentile)
IWA-NF-002	99.9% availability

4. Features Not to be Tested

- Core actuarial engine internals (vendor-tested)
- Third-party KYC provider internals
- Payment gateway internals
- Cloud infrastructure (AWS/Azure) internals

5. Test Approach / Strategy

Levels:

- Unit tests (module-level, JUnit + Jest)
- Integration tests (service ↔ service, service ↔ DB, service ↔ Kafka)
- System tests (end-to-end workflows)
- Acceptance tests (UAT with business users)

Types:

- Functional testing (quotes, policies, claims)
- Regression testing (automated via CI/CD)
- Performance testing (JMeter / k6)
- Usability testing (WCAG 2.1 AA)
- Security testing (OWASP ZAP, pen testing)

Entry Criteria: Stable build deployed to test env, test data ready, test cases reviewed.

Exit Criteria: 100% planned test cases executed, 0 critical defects open, all acceptance criteria met.

5.1 Security Validation

- Validate PII masking in logs and UI
- TLS 1.2+ verification (SSL scan)
- PCI-DSS & GDPR compliance checks
- Fuzzing on quote/claim input fields
- Penetration testing on auth and workflow APIs
- RBAC bypass attempts

6. Test Environment

Hardware: Cloud VMs (Kubernetes cluster), HSM for crypto tests.

Software: IWAS v1.0, PostgreSQL test DB, Kafka test cluster, Camunda engine, S3 test bucket.

Tools: Selenium / Playwright (UI), Postman / RestAssured (API), JMeter / k6 (performance), Jira (defects), GitHub Actions (CI/CD).

Test Data: Synthetic customer profiles, dummy policies, mock claim documents, test payment cards.

7. Test Schedule

Milestone	Date
Test case design	05-Sep-2025
Environment setup	07-Sep-2025
Test execution start	08-Sep-2025
Test execution end	20-Sep-2025
UAT	22-Sep-2025 to 25-Sep-2025

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated)
- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports

- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
Project Lead	Sanjith Suresh Kumar	Overall ownership, sign-off, coordination
QA Lead	Sakaray Vishal	Prepare plan, coordinate execution, defect triage
Developer Lead	Samarth A	Support defect fixes, code review, smoke builds
DevOps / Security Engineer	Sampreeth Jaladi	CI/CD pipeline, environment setup, security testing

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds from dev team
Test environment downtime	Maintain backup environment on cloud VM
Dependency on third-party KYC / payment vendors	Engage vendor early, maintain test stubs/mocks
BPMN workflow complexity causing test flakiness	Use deterministic test workflows + versioned BPMN files

11. Assumptions & Dependencies

- KYC, payment, and actuarial sandboxes will be stable and available
- Test data (customers, policies, claims) will be provided before execution
- CI/CD pipeline will be operational before test execution start

12. Suspension & Resumption Criteria

Suspend testing if:

- Environment unavailable for >4 hours
- Build too unstable (blocks >30% test cases)
- Critical security vulnerability discovered

Resume testing if:

- Blocking defects resolved
- Environment stabilized
- Security patch applied and verified

13. Test Case Management & Traceability

RTM ensures mapping of SRS requirements to test cases. Example:

- IWA-F-001 (Quote generation) → TC-Quote-01, TC-Quote-02
- IWA-F-010 (Policy issuance) → TC-Policy-01, TC-Policy-02
- IWA-F-020 (Claim registration) → TC-Claim-01, TC-Claim-02
- IWA-F-021 (BPMN workflow) → TC-WF-01, TC-WF-02
- IWA-NF-001 (Response time) → TC-Perf-01
- IWA-SR-001 (TLS) → TC-Sec-TLS-01

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed/failed
- Defect density
- Defect aging
- Requirement coverage
- Automation coverage

Reports:

- Daily execution status (shared via dashboard)
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
Project Lead	Sanjith Suresh Kumar	—
QA Lead	Sakaray Vishal	—
Dev Lead	Samarth A	—
DevOps / Security	Sampreeth Jaladi	—